Algebraic manipulation

1 Solve the following equations:

a $x^2 - 25 = 0$

b $x^2 = 98$

c $81x^2 - 64 = 0$

d $5\left(p^2 - 3\right) = 705$

e $x\left(7x - 12\right) = 0$

f $\left(x - 4\right)^2 = 10$

g $\left(4x + 3\right)^2 = 64$

h $\left(4x - 9\right)^2 = 0$

i $\left(3 - 3x\right)^2 = 0$

j $\frac{m}{6}\left(m - 10\right) = 0$

2 Solve the following equation for x, in terms of a and c. Assume a and c are positive.

$$ax^2 - c = 0$$

Factorisation

3 Solve the following equations by factorising:

a $x^2 + 12x = 0$

b $3y - 15y^2 = 0$

c $m^2 = 14m$

d $x^2 + 19x + 90 = 0$

e $m^2 = m + 20$

f $x^2 - 20x + 100 = 0$

g $-x^2 + x + 20 = 0$

h $m^2 - 27m = -182$

i $\frac{x^2 - 5x}{8} = 3$

j $\left(y + 1\right)^2 = 4y + 4$

k $4x^2 + 8x - 32 = 0$

l $-4x^2 + 25x - 36 = 0$

m $4x^2 + 17x + 15 = 0$

n $5x^2 + 14x + 8 = 0$

o $4x^2 - 17x + 15 = 0$

p $3x^2 - 7x - 20 = 0$

q $\frac{2x^2 - 19x}{3} = 20$

r $15 - 11b - 12b^2 = 0$

s $6k^2 = -7 + 13k$

4 On the graph of $y = x^2 - 4$, there are two points where $y = 12$. What are the x-coordinates of these two points?

5 The equation $x\left(x - 7\right) = -10$ has a positive integer solution of $x = 5$. What is the other solution to the equation?

6 Write a monic quadratic equation given the two solutions. Leave the equation in factorised form.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

a $x = 7$ and $x = -4$

b $x = 0$ and $x = -1$

7 Write a monic equation given the two solutions: $x = -3$ and $x = -5$. Write the equation in expanded form.

Using technology

8 Solve the following equations by using technology. Round your answers to 1 decimal place where necessary.

a $3x^2 - x - 10 = 0$

b $x^2 + 2x - \frac{21}{4} = 0$

c $4.6x^2 + 7.3x - 3.7 = 0$

9 Consider the equation $3x^2 = 6$.

a Solve the equation by using technology, giving your answers in exact form.

b Give the solutions as decimals rounded to the nearest tenth.

10 Consider the equation $0 = -x^2 + 2x - 1$. Use technology to solve the equation for x.

a Write down how many solutions the equation has.

b Hence, state how many x-intercepts there are on the graph of the function

$$y = -x^2 + 2x - 1$$

Applications

11 How many real solutions does the equation $x^2 + 64 = 0$ have?

12 Solve the following equation for x. Leave your answer in surd form.

$$\frac{3x + 1}{3x - 1} - \frac{3x - 1}{3x + 1} = 5$$

13 Solve $\left(5x^2 + 13x + 6\right)\left(2x^2 + 13x + 20\right) = 0$.

14 The equation $4x^2 + kx + 16 = 0$ has one solution: $x = 2$. What is the value of the coefficient k?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

15 Consider the quadratic expression $ax^2 - 32x - 80$. If you substitute $x = 4$ into the expression, what must be the value of a to make the expression equal 0?

16 The Widget and Trinket Emporium has released the forecast of its revenue over the next year. The revenue R (in dollars) at any point in time t (in months) is described by the equation

$$R = -(t - 18)^2 + 16$$

Solve the equation $-(t - 18)^2 + 16 = 0$ to find the times at which the revenue will be zero.

17 Software engineers are designing a self-serve checkout system for a supermarket. They notice that the traffic through the store during the day is described by the function

$$C = -t(t - 12)$$

where C is the number of customers and t is the number of hours after the store opens. To meet the peak demand, the engineers allow for an extra checkout machine to automatically turn on when the number of customers first reaches 32 people, and to automatically turn off when it next falls below 32 people.

a Find the times t when the number of customers is equal to 32 people.

b Hence, after how many hours will the the extra checkout machine turn on after opening?

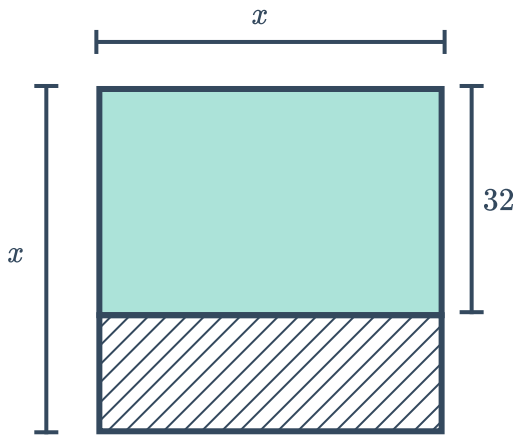
c How many hours will the extra machine be on for?

18 Homer needs a sheet of paper x cm by 32 cm for an origami koala. The local origami supply store only sells square sheets of paper.
The lower portion of the image below shows the excess area A of paper that will be left after Homer cuts out the x cm by 32 cm piece. The excess area, in cm^2, is given by the equation

$$A = x(x - 32)$$

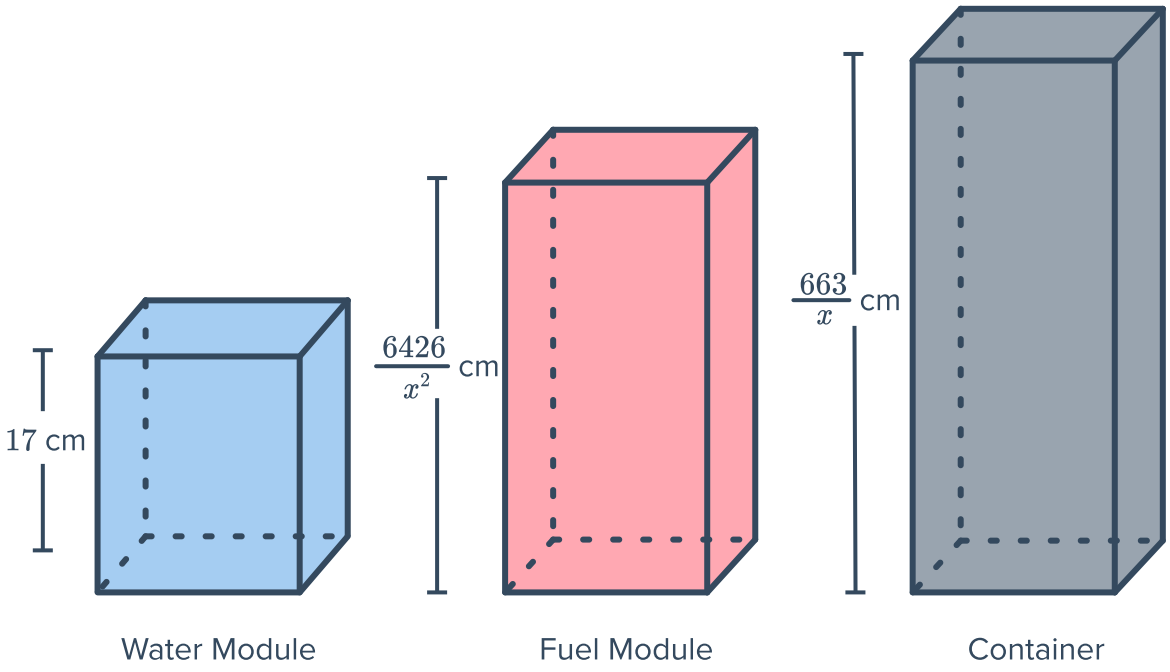
a At what lengths x will the excess area be zero?

b For what value of x will Homer be able to make an origami koala with the least amount of excess paper?



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

19 An interplanetary freight transport company has won a contract to supply the space station orbiting Mars. They will be shipping stackable containers, each carrying a fuel module and a water module, that must meet certain dimension restrictions.
The design engineers have produced a sketch for the modules and container, shown below. The sum of the heights of both modules equal to the height of the container.



a Write an equation that equates the height of the container and the sum of the heights of the modules.

b Find the possible values of x.

c Find the tallest possible height of the container. Give your answer to two decimal places.

20 Executives at the Widget Emporium are discussing whether to merge their company with the Trinket Bazaar, a large competitor.
Market analysis shows that the extra revenue the company will receive can be modelled by the equation $R = 0.25t^2$, and the extra costs by the equation $C = 3.5t$. R and C are measured in thousands of dollars and t is measured in months after the merger.

a Use the solve function on your CAS calculator to find the times at which the extra revenue R will match the extra cost C.

b The executives decide that they can only afford to operate at a loss for one year. Based on this requirement, would you advise that the Widget Emporium merge with the Trinket Bazaar?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647